PROJECT SYNOPSIS

AI-BASED DISASTER EARLY WARNING AND RESPONSE SYSTEM

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Project Declaration

We hereby declare that the synopsis entitled **“AI-Based Disaster Early Warning and Response System”** is our original work and has not been submitted to any other University or Institution for the award of any degree or diploma.

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Introduction & Motivation

India, especially the Himalayan state of **Uttarakhand**, is frequently hit by devastating floods, flash floods, and landslides.

Current disaster information is fragmented across multiple agencies (IMD, NASA, GSI) and formats, creating significant delays for local decision-makers.

> **Project Motivation:** Use AI to unify environmental data into a single, map-based interface for regional disaster preparedness and response.



Vulnerable regions in the Himalayan state of Uttarakhand.

Problem Statement & Objectives

CORE OBJECTIVES

- ✓ Collect and prepare weather, terrain, and historical hazard data.
- ✓ Develop flood and landslide risk models using Random Forest.
- ✓ Build a React dashboard with an interactive Leaflet risk map.
- ✓ Develop FastAPI endpoints for prediction and alert history.

CRITICAL SCOPE & LIMITATIONS

- Prototype status: Not a replacement for official warning services.
- Data Dependency: Performance relies on the quality of available labels.
- Target: Preliminary preparedness, not tactical emergency response.

Literature Survey & Research Gap

Category	Sources / Methods	Key Findings
Environmental Data	IMD, NASA (GPM/SRTM), GSI	Satellite precipitation and elevation data provide critical features for mountainous terrain.
Risk Assessment	Random Forest, Logistic Regression	Random Forest effectively handles non-linear relationships in slope and rainfall data.
Visualisation	Leaflet, OpenStreetMap	Real-time map-based dashboards significantly improve situational awareness.

The Identified Gap

While many systems exist for individual hazards, there is a need for a **unified decision-support prototype** that integrates multi-hazard risk assessment, map-based visualization, and a formal alert history specifically for the Himalayan context.

Proposed Methodology

01

Data Intake

IMD, NASA, GSI APIs & historical hazard records

02

ML Inference

Random Forest prediction models (Proposed)

03

API Layer

FastAPI endpoints & PostgreSQL storage

04

Dashboard

React frontend with Leaflet interactive map

System Layer	Primary Responsibility
Data & Processing	Cleaning, feature engineering, and inference pipelines.
Backend & DB	API routes, alert rules, and persistent prediction logs.
Frontend	Real-time visualization, charts, and alert management.

Technologies Used

FRONTEND

React.js

Component-based UI with Leaflet for maps and Axios for API calls.

BACKEND

FastAPI

Python

Asynchronous endpoints, Pydantic validation, and alert logic.

DATA & ML

scikit-learn

Pandas

Data cleaning, Random Forest models, and model evaluation.

INFRASTRUCTURE

PostgreSQL

Git

Persistent data storage, SQLAlchemy ORM, and version control.

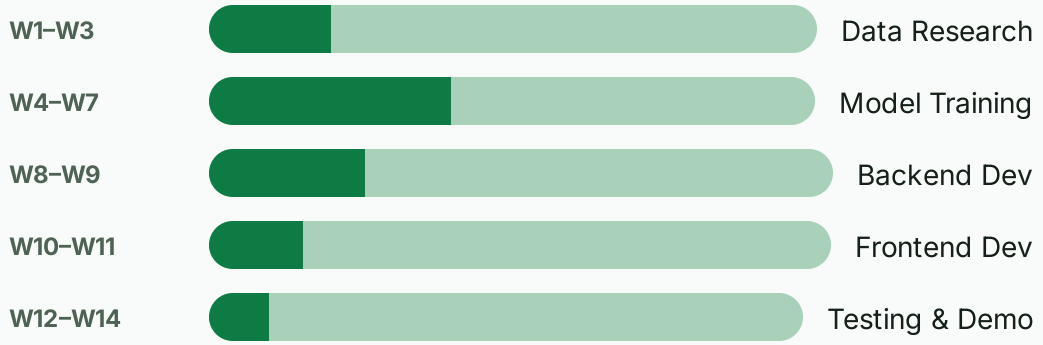
Evaluation & Timeline

PERFORMANCE METRICS

Target	Metric (Proposed)
Risk Models	Precision, Recall, F1-Score
System API	Response Time, Uptime
Alert Pipeline	Delivery Success Rate

* Leakage-aware test splitting will be used for model validation.

14-WEEK TIMELINE



Major Contributions & Future Scope

INTENDED CONTRIBUTIONS

- → A unified full-stack prototype (React + FastAPI) for Himalayan hazards.
- → Hazard-specific ML pipelines for flood and landslide risk.
- → Map-based visualization of risk categories and data timestamps.

FUTURE EXTENSIONS

- → Integration of real-time sensor networks for higher local precision.
- → Expansion to other hazards (e.g., forest fires, seismic activity).
- → Mobile-native alerts using push notifications and SMS gateways.

Conclusion

- Proposed as a **web-based decision-support prototype** for flood and landslide risk in Uttarakhand.
- Integrates data processing, machine learning, and map-based visualization.
- Focuses on **preliminary preparedness** and communication of uncertainty.

> *"This prototype is intended for academic learning and preparedness support; it does not replace or override official disaster warning services."*

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